



National University

of Computer & Emerging Sciences Peshawar Campus

Student Name: _____

Roll Number: _____

Semester: Fall 2025

Exam: Mid

Marks: 30

Weightage: 20 to 25

Time Allowed: 90 min

Date: 15, Oct, 2025

Course: CL-2001 Data Structure Lab

Instructor: M Saad Khan

Paper Type: Even

Note:

- Attempt all questions.
 - Understanding the question is the part of Exam.
 - Create a folder with the format (RollNo_yourname). Include all the C++ files(23P0679_Q1.cpp) in that folder. Use the following format for naming the files (RollNo_QuestioNo.cpp).
 - An individual may be assigned a straight-forward 0 if the submitted assessed task is copied/cheated from another individual/internet.
 - Comments your Code Properly. Use meaning full variable names for each and every question.
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Problem: 1

Marks: 07

Write a C++ program to check whether the array is in ascending order, descending order, or not in any order.

Note: Use only pointers to access array elements. Index-based solutions are not allowed; otherwise, zero marks will be awarded.

E.g : array {1, 5, 6, 7} The array is an ascending order.

Problem: 2**Marks: 15**

Write a C++ program to simulate a Customer Service Call Center using a queue implemented with arrays.

Your program should:

1. Ask the user how many customers can wait in the call queue (hold line) before an overflow message is triggered.
2. Provide a menu with the following options:
 - Add a new caller to the queue (customer calls/enqueue).
 - Connect the next caller to an agent (remove from queue/dequeue).
 - See who is next to be connected (peek).
 - Display all waiting callers in the queue.
 - Exit the program.
3. Ensure that the queue follows the FIFO (First In, First Out) principle.

Note: Passenger class must have attributes name and issueType e.g (“Billing”, “Technical”, “Sales”).

Problem: 3**Marks: 08**

Write a C++ program that takes a target value. After taking the target value, the singly linked list should be reversed from that value onward.

For example:

If the linked list is 2->4->6->1->9 and the target is 6, then the linked list will become 6->1->9->2->4.

If the target is 9, then the output linked list will be 9->2->4->6->1.